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Co2 - AT3

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SIMATS ENGINEERING
ASSESSMENT TOOL

COURSE NAME: Theory of Computation COURSE CODE: CSA1301

COURSE OUTCOME COVERED

CO2: Deign Finite Automata DFA, NFA, ε-NFA and demonstrate their equivalence for recognizing regular languages. (BL:3)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks:20**
Duration : 1 Hour

Scenario Based

Code of Conduct:

I C.Manohar 192372246 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

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Signature of the student:

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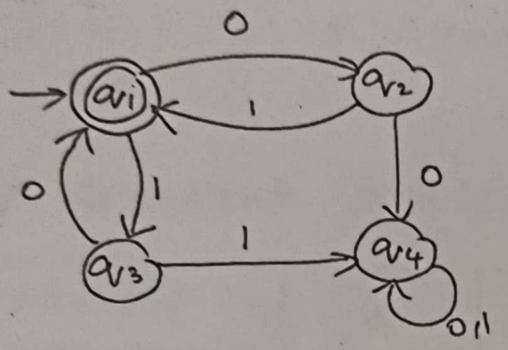
1.Given a DFA used for validating a specific PIN pattern bbbc, bca ,c, ca, caaaaaa minimize the DFA and explain how state reduction improves performance and efficiency in embedded systems

2.Demonstrate how a given finite automaton can be converted into a regular expression



3.Consider the following grammar

S->aSc|T|U|e

T-> bTc|e

U->aUb|e

A} What is the language defined by the grammar?

B)Give the derivation tree for the word aaabbc

C Grammar is Ambiguous if there exists two distinct derivations for a word. Is the above grammar ambiguous? Justify.

4.I)Show that if L1 and L2 are regular then (L1+l2) also regular

Ii)Is the language { a^p| p is prime} regular? Justify

Step 1:

Transitions for the five strings:

1. bbbc
2. bca
3. c
4. ca
5. caaaaaa

A convenient DFA is:

State/IP	a	b	c	Final
q_0	D	q_1	q_7	NO
q_1	D	q_2	q_5	NO
q_3	D	q_3	D	NO
q_5	D	D	F	NO
q_7	F	D	D	NO
q_8	q_8	D	D	yes
q_9	q_9	D	D	yes
q_{10}	q_{10}	D	D	No
q_{11}	q_{11}	D	D	NO
q_{12}	q_{12}	D	D	yes
q_{13}	D	D	D	yes
F	D	D	D	yes
D	D	D	D	

⇒ F represents the common accepting terminal state.

⇒ D is the dead/trap state.

The minimized DFA has 12 states including the dead state.

State reduction improves:

1. Memory usage
2. processing speed
3. power efficiency

(2)

Start / Final state : q_1

→ Transitions :

$q_1 \xrightarrow{0} q_1$

$q_1 \xrightarrow{0} q_2 \xrightarrow{1} q_1$

$q_1 \xrightarrow{1} q_3 \xrightarrow{0} q_1$

$q_2 \xrightarrow{0} q_4$

$q_3 \xrightarrow{1} q_4$

The path that return to the final State q_1 are:

0, 01, 10

$\boxed{R = (0+01+10)^*}$ ✓

(3) a]

For U :

$U \to aUb | \epsilon$

$L(U) = \{a^n b^n | n \geq 0\}$

Ex :- ϵ, ab, aabb....

For T :

For T:

$T \rightarrow bTc / \epsilon$

$L(T) = \{ b^n c^n / n \geq 0$

Ex:- ϵ, bc, bbcc..

Form 1: using 'T'

$S \Rightarrow a^k T c^k$

Since T provides $b^n c^n$

$a^k b^n c^{n+k}$

$L_1 = \{ a^k b^n c^{n+k} / k, n \geq 0 \}$

Form 2: using 'U'

$S \Rightarrow a^k U c^k$

$a^{k+n} b^n c^k$

$L_2 = \{ a^{k+n} b^n c^k / k, n \geq 0 \}$

$$\boxed{L(G) = \{ a^k b^n c^{n+k} / k, n \geq 0 \} \cup \{ a^{k+n} b^n c^k / k, n \geq 0 \}}$$

b]

$S \rightarrow \cancel{aSc}$

$S \rightarrow aUc \quad S \rightarrow U$

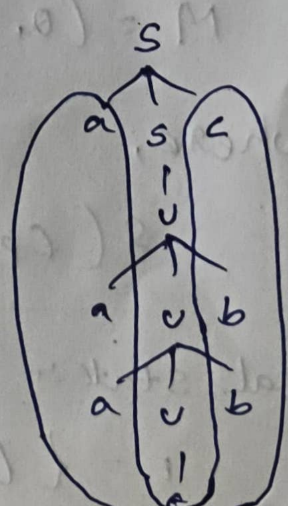
$S \rightarrow \underline{aaUbc} \quad U \rightarrow aUb$

$S \rightarrow aa\,\underline{aUb}\,bc \quad U \rightarrow aUb$

$S \rightarrow aaabb\,c \quad U \rightarrow \epsilon$

∴ Given string "aaabbc" is accepted

Parse Tree:



c] Derivation - 1: (LMD):

$S \rightarrow aSc$

$S \rightarrow a\epsilon c \rightarrow S \rightarrow \epsilon$

$S \rightarrow ac$

$\therefore$ The grammer is ambiguous

Derivation - 2:

$S \rightarrow aSc$

$S \rightarrow awc$

$S \rightarrow ac.$

4) i] $M_1 = (Q_1, \Sigma_1, \delta_1, q_1, F_1)$ and

$M_2 = (Q_2, \Sigma_1, \delta_2, q_2, F_2)$

$L(M_1) = L_1$ & $L(M_2) = L_2$

New DFA :-

$M = (Q_1 \times Q_2, \Sigma_1, \delta_1 (\cancel{q_1} q_2), F)$

where,

$\delta((p, q), a) = (\delta_1(p, a)\ \delta_2(q, a))$

Final state :-

$F = \{ (p, q) \mid p \in F_1 \text{ OR } q \in F_2 \}$

Hence

Hence: $L(M) = L_1 \cup L_2$

$L_1 \cup L_2$ is Regular

ii) <u>Step 1:</u>

Assume 'L' is Regular

$L = \{a^p / p \text{ is prime}\}$

a^p belongs to L

where:-

$|xy| \leq m$

$|y| > 0$

$|y| = k$

$1 \leq k \leq M$

step. 2:-

$i = p+1$

$xy^{p+1}z$

$p + pk$

<u>factorizing:</u>

$p + pk = p(1+k)$

$p > 1$

$1 + k > 1$

$p(1+k)$

$xy^{p+1} z \notin L$

$$\boxed{L = \{a^p / p \text{ is prime}\} \text{ is not regular}}$$

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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

92/100